B.Sc. Engg. CSE 6th Semester

23 February 2023 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION | **SUMMER SEMESTER, 2021-2022**

DURATION: 1 HOUR 30 MINUTES | **FULL MARKS: 75**

CSE 4619: Peripherals and Interfacing

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) Write short notes on fixed and variable addresses. Explain the memory-mapped I/O concept using the fixed addressing scheme. — 5 + 5 (CO1) (PO1)

   b) What is Harvard Architecture for microcontroller design? Explain the use and functions of appropriate pins of ATMega16 microcontroller for analog I/O and external clock input operation. — 3 + 7 (CO1) (PO1)

   c) List out and describe the operations of at least two applications which use a microcontroller. — 5 (CO1) (PO1)

2. a) If an analog data is converted into digital format, then is it possible to retrieve the same analog data from that digital one? Justify your answer with appropriate reason. — 10 (CO3) (PO1)

   b) Suppose, it is given $V_{in} = 0.5\ Volt$, $V_{ref} = 1\ Volt$ and 12-*bit* of resolution for a *Successive Approximation A/D* converter. Find the 12-*bit* digital output for the given V_{in}. Also find the analog value for that digital output using the *Weighted Sum D/A* conversion method. — 5 + 5 (CO2) (PO1)

   c) How can a 8259 PIC handle 64 Interrupt levels? Explain with necessary diagram. — 5 (CO1) (PO1)

3. a) Suppose, an 8086 microprocessor is asked to address the 32nd and 64th 8255 sequentially and to write a control word at the control register of that 8255. Consider, Port-A is in Mode-2, Port-B is in Mode-1 as an output port. Now, derive the sequential binary values of A7 – A0 pins connected to 8255 and draw the control word format for 8255. — 5 + 5 (CO4) (PO2)

   b) Write the basic differential features between 8155 and 8255 Programmable Peripheral Interface. — 10 (CO1) (PO1)

   c) Draw the timing diagram of a 8255 PPI when its Port-A is used as input port having interrupt request with a 8-bit data of all '1's and Port-B is used as output port having no interrupt requested with a 8-bit data of all '0's; If Port-C is in the handshaking mode then also derive the command register value. — 5 (CO2) (PO2)